# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

# CHEMISTRY

Past Paper 2024 • Paper Code: PR–XI (A-I-A) (24)

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** PR–XI (A-I-A) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Better cooking in Pressure Cooker is an example of:

(A) Reduced P
(B) Increased P
(C) Constant P
(D) None

Q2. When coordination number of a crystalline substance is 0.54, the structure will be:

(A) Trigonal
(B) Tetrahedral
(C) Octahedral
(D) Cubic

Q3. The nature of bonds between atoms in graphite is:

(A) Ionic
(B) Covalent
(C) Molecular
(D) Metallic

Q4. Which of the given representation is incorrect form of law of mass action?

(A) Kc = [Product]
(B) Kc x [Product]
(C) Kc × [Product]/[Reactant]
(D) None of these

Q5. Due to common ion effect, the solubility of a weak electrolyte is always ____ by addition of strong electrolyte.

(A) Enhanced
(B) Suppressed
(C) No change
(D) None

Q6. Conjugate acid-Base pairs are always differ by:

(A) $1H^+$ (B) $2H^+$

(C) $3H^+$ (D) $4H^+$

Q7. The pH of the blood is:

(A) 6.4 (B) 7.0

(C) 7.4 (D) 8

Q8. In ice cream, sugar is:

(A) Solute (B) Mixture

(C) Solvent (D) Dispersion phase

Q9. ΔH can be measured indirectly by applying.

(A) Gas Laws (B) Avogadro's Law

(C) Hess's Law (D) Faraday's Law

Q10. What is the mass of 0.5 moles of CaCO3?

(A) 100g (B) 75g

(C) 50g (D) 25g

Q11. Which of the given has highest percentage of nitrogen?

(A) (NH4)2SO4 (B) NH4Cl

(C) NH4NO3 (D) (NH4)3PO4

Q12. What is the energy of an election in 1st orbit of H-atom?

(A) -1312.36 KJ/mol (B) -1012.36 KJ/mol

(C) -912.36 KJ/mol (D) -427.36 KJ/mol

Q13. $\sqrt{\upsilon} \times Z$ is the mathematical form of:

(A) Avogadro's Law (B) Moseley's Law

(C) Boyle's Law (D) Graham's Law

Q14. The bond angle in NH3 molecule is:

(A) 109.5° (B) 107.5°

(C) 104.5° (D) 101.5°

Q15. Total number of sigma bond in ethylene molecule is/are:

(A) 1 (B) 3

(C) 5 (D) 6

Q16. All are correct except:

(A) 1 atm=760 torr
(B) 760 torr=760 mm Hg
(C) 760 torr=101325 Pa
(D) 1 atm=1.01325 KPa

Q17. The spreading of perfume in air is due to:

(A) Diffusion
(B) Effusion
(C) Low density
(D) Attraction with air

Q18. The B.P of water is greater than B.P of H2S. this is due to the presence of:

(A) Hydrogen Bonding
(B) Dipole-dipole interaction
(C) London dispersion forces
(D) All these

## SECTION-B

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Attempt any TEN parts. Each part carries FOUR marks.

i) ΔH for solids and liquids become equal to ΔE. Why?

ii) Calculate E° cell from the following half cell reaction: (a) $Zn \rightarrow Zn^{2+} + 2e-$ ($E^{o}ox = 0.76V$) (b) $Cu^{2+} + 2e- \rightarrow Cu$ ($E^{o}Red = 0.34V$)

iii) What are state of the system and state functions?

iv) Explain mole fraction?

v) Write down characteristics of the catalyst?

vi) Explain why the sum of PKa and PKb is always equal to 14?

vii) What is meant by buffer action?

viii) What is the role of a catalyst to affect the position of equilibrium?

ix) Why density of ice is low as compared to water?

x) Why the droplet of water is spherical?

xi) How the existence of the hydrogen bonding can be proved?

xii) Calculate the density in $g/dm^3$ of NH3 at 25°C and 1.2 atm?

xiii) Give the reason that why O2 molecule is paramagnetic?

## SECTION-C

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 27

Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q14.

(a) What is the %age yield of copper oxide when 24.8 g of copper carbonate ($CuCO_3$) is heated to produce 13.9g of copper oxide (CuO).

(b) Write down the characteristics of neutron?

Q15.

(a) Explain Magnetic and spin quantum numbers in detail.

(b) Explain elevation in boiling point of liquids?

Q16.

(a) Explain Graham's Law of diffusion and effusion.

(b) Balance the following equations by oxidation number method. (i) $Zn + HNO_3 \rightarrow Zn(NO_3)_2 + NO + H_2O$ (ii) $Fe + V_2O_3 \rightarrow Fe_2O_3 + VO$

Q17. Write note on the following. (a) Partially Miscible Liquids (b) Electrolytic Cell (c) Boyle's Law